

Educational Building HVAC-BMS Digital Twin

HVAC-BMS Reduced-Order Digital Twin Report

KPI	Value	Unit
Simulation strategy	S3	-
Simulation duration	2.000	days
Electric HVAC energy	3056.152	kWh
Thermal/boiler input	0.000	kWh
Peak electric demand	152.371	kW
Average chiller COP	6.025	-
Comfort-degree hours	0.587	C.h
Occupied discomfort ratio	3.530	%
Maximum average CO2	1044.974	ppm
Electricity cost	366.738	USD
Operational carbon	1375.269	kgCO2
Alarm events	26.000	events
Final degradation index	0.042	0-1

BMS alarm log

Timestamp	Code	Severity	Message	Value
2026-07-01 10:30:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1018.3596980733864
2026-07-01 11:00:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1030.7529196581775
2026-07-01 11:30:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1037.4126479149547
2026-07-01 12:00:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1040.9659922369867
2026-07-01 12:30:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1042.8515297272045
2026-07-01 13:00:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1043.8496768427226
2026-07-01 13:30:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1044.3786657404542
2026-07-01 14:00:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1044.660336501766
2026-07-01 14:30:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1044.8115395025525
2026-07-01 15:00:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1044.8936219888744
2026-07-01 15:30:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1044.9388059527853
2026-07-01 16:00:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1044.964083288883
2026-07-01 16:30:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1044.9737421883895
2026-07-02 10:30:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1018.9885424534335
2026-07-02 11:00:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1031.0842995481341
2026-07-02 11:30:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1037.5870292532252
2026-07-02 12:00:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1041.0578997083196
2026-07-02 12:30:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1042.9002006566259
2026-07-02 13:00:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1043.8756582932117
2026-07-02 13:30:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1044.3926900429274
2026-07-02 14:00:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1044.6680127911375
2026-07-02 14:30:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1044.8158105015202
2026-07-02 15:00:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1044.8960422522825
2026-07-02 15:30:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1044.9402047882438
2026-07-02 16:00:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1044.9649085974904
2026-07-02 16:30:00	BMS-IAQ-CO2	High	Average occupied-zone CO2 exceeded the configured limit.	1044.9743453957003